

Adversary-Grounded Location Privacy and a Density-Adaptive Hybrid Mechanism for IoT Smart Cities

Pushkal Gupta, Praagya Garg, M. Nagasai Dattu, and S. Ebenezer Juliet

Abstract—Smart cities depend on continuous location reporting from IoT devices, yet the privacy mechanisms that protect these reports are evaluated in isolation, with incompatible datasets, metrics, and threat models, so no fair comparison exists. We make two contributions that go beyond benchmarking. First, we replace the ad-hoc privacy scores common in the literature with an adversary-grounded privacy metric: privacy is the expected error of an optimal Bayesian adversary, measured in metres and therefore directly comparable across mechanism families. Crucially, we evaluate every mechanism under two threat models—a single-observation snapshot adversary and a Viterbi trajectory-tracking adversary equipped with a Markov mobility prior—and show that a single privacy scale is inadequate: mechanisms strong against one adversary can be weak against the other, and the two adversaries induce different rankings. Second, we introduce DA-Hybrid, a density-adaptive mechanism that combines bounded graph k -anonymity with a graph-constrained differential-privacy fallback whose budget strengthens in sparse regions; it guarantees 100 % availability and on-graph outputs while achieving the lowest spatial error among all full-availability mechanisms. We evaluate five classical mechanisms and DA-Hybrid on a real central-Beijing OpenStreetMap road network under two real mobility datasets (GeoLife and T-Drive), together with a radio-technology sensitivity analysis of the energy model. Rankings under the adversary metric are shown—not merely asserted—to be stable from the grid abstraction to the real network, while service availability is found to be dataset-density-driven rather than mechanism-intrinsic. The complete framework is released for reproducible extension.

Index Terms—Location privacy, k -anonymity, differential privacy, graph-based privacy, Bayesian adversary, trajectory privacy, Internet of Things, smart cities.



1 Introduction

The proliferation of Internet of Things (IoT) devices in smart cities produces an unprecedented volume of fine-grained spatiotemporal data. Smartphones, connected vehicles, and environmental sensors continuously report location information that enables urban analytics, traffic management, and emergency response, yet simultaneously exposes home and workplace locations, health-facility visits, and detailed mobility, even after identifiers are removed [1].

Two decades of research have produced diverse mechanisms—spatial k -anonymity [2], differential privacy [3], density-aware adaptation [11], graph-constrained perturbation [10], and temporal cloaking [2]. But each is typically evaluated on its own dataset, with its own metric, under its own threat model, so a practitioner cannot obtain a defensible answer to “which mechanism should I deploy?” A first response is to place all mechanisms on a shared substrate. We argue that this is necessary but not sufficient: the privacy axis itself must be made rigorous and threat-model-aware, and the empirical study must run on real road networks and more than one dataset before its conclusions can be trusted. This paper does both, and additionally contributes a new

mechanism that the rigorous evaluation shows to occupy an otherwise-empty operating point.

Contributions.

- 1) An adversary-grounded, threat-model-aware privacy metric (Section 4). We define privacy as the expected localisation error of an optimal Bayesian adversary [5], instantiated as (i) a snapshot adversary and (ii) a Viterbi trajectory adversary using a first-order Markov mobility prior estimated from data. This yields a single physical unit (metres) comparable across families, and it exposes that the two threat models rank mechanisms differently—evidence that the widely used single privacy scale is invalid.
- 2) DA-Hybrid, a new density-adaptive mechanism (Section 6). Per report, DA-Hybrid uses the local user density to choose bounded graph k -anonymity where density permits and a graph-constrained DP fallback (with density-adaptive ϵ) otherwise. It gives a per-report formal guarantee everywhere, 100 % availability by construction, and on-graph outputs, and it attains the lowest spatial error among all full-availability mechanisms.
- 3) A real-network, multi-dataset, energy-sensitive evaluation (Sections 7–8). We evaluate on a real central-Beijing OpenStreetMap road graph under both GeoLife and T-Drive, and sweep the energy model across BLE/Zigbee/LoRa/NB-IoT/LTE-M radios, separating robust conclusions from radio- or granularity-

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specific ones.

- 4) An open, extensible framework releasing all mechanisms, adversaries, datasets, and analysis for reproducibility.¹

2 Related Work

Spatial cloaking and k -anonymity. Sweeney [1] defined k -anonymity; Gruteser and Grunwald [2] introduced spatial cloaking, and Mokbel et al. [6] and Duckham and Kulik [7] formalised region growth on spatial structures and connected subgraphs. Recent work extends cloaking for edge computing and trajectory publication [18], but assumes Euclidean space and fixed parameters and is rarely compared across families.

Differential privacy for location. The Laplace mechanism [4] and geo-indistinguishability [8], [9] add calibrated noise to coordinates; Xiao and Xiong [12] showed that temporal correlation degrades these guarantees. Unconstrained noise can place reports off the road network, motivating graph projection [10], [13].

Trajectory privacy and evaluation. Chow and Mokbel [14] formalised trajectory privacy; recent LDP synthesis [15], [16] and generative approaches [17] advance within single families. Crucially, the standard way to measure location privacy is the expected error of an optimal adversary [5]—the view we adopt here in place of family-specific scores. Prior comparative studies fix a dataset or a metric but not the threat model, and none, to our knowledge, evaluate a common mechanism set under both a snapshot and a trajectory adversary on a real road network across two datasets, which is the gap this paper fills.

3 System Model and Threat Models

3.1 Graph-Based Spatial Model

The urban environment is an undirected weighted graph $G = (V, E)$ where vertices are spatial locations and edges are road connectivity, with edge weights the metric road distance. We use two graphs over central Beijing (latitude [39.8, 40.1], longitude [116.2, 116.5]): a controlled 30×30 grid (900 nodes) that removes topological confounds, and a real OpenStreetMap drive network for the same region, consolidated to 3,154 nodes and 5,111 edges with irregular degree (mean 3.24, maximum 23), dead-ends (16%), and hierarchy—properties absent from the grid. The canonical distance is the shortest path; distances are shared across all mechanisms so no mechanism is advantaged by a different metric.

3.2 Trajectory Data

We use two real datasets. GeoLife [19], [20] contains GPS traces from 182 users in Beijing (2007–2012); after bounding-box filtering and a 200 km/h speed filter it yields ~ 16.9 M records on the grid and ~ 11.4 M map-matched to the real graph. T-Drive is a denser one-week trace of $\sim 10,000$ Beijing taxis, yielding ~ 4 M records map-matched to the same real graph. Using two datasets with very different density is essential: it separates genuine mechanism properties from single-dataset quirks (Section 8).

3.3 Two Threat Models

Privacy is meaningful only against an adversary. We consider two, so that mechanisms defending different threats are each judged against the threat they target:

- a snapshot adversary that observes a single released report and infers the true node; and
- a trajectory adversary that observes a sequence of reports for a user and reconstructs the trajectory using a mobility model.

Both possess public background knowledge estimated from the data: a population prior $\pi(v)$ (node-occupancy distribution) and a first-order Markov mobility model $P(v_{t+1} | v_t)$. The adversary knows the mechanism but not its private randomness. Section 4 makes both operational.

4 Adversary-Grounded Privacy Metric

Comparative studies commonly reduce privacy to a mechanism-specific score in $[0, 1]$ —e.g. $k/k_{\max}$ for k -anonymity and a log-scaled ϵ for DP. Such scores have no common operational meaning and cannot answer whether $k=3$ anonymity hides a user better than $\epsilon=1$ geo-indistinguishability. We instead define privacy as the expected error of an optimal adversary that inverts the mechanism using public knowledge [5], measured in metres and therefore comparable across all mechanisms.

4.1 Snapshot Adversary

Given one release o , the adversary forms the posterior $P(v | o) \propto P(o | v) \pi(v)$ over the true node v and outputs the Bayes estimate

$$\hat{v} = \arg \min_{v' \in V} \sum_{v \in V} P(v | o) d(v', v), \quad (1)$$

minimising expected graph distance. Privacy is $\text{AE}_{\text{snap}} = \mathbb{E}[d(v_{\text{true}}, \hat{v})]$ in metres. The observation model $P(o | v)$ differs by family—for cloaking mechanisms it restricts the prior to the released anonymity set; for the planar-Laplace mechanisms it is the Laplace emission density times the prior—but the measured quantity is identical for all.

4.2 Trajectory Adversary

A stronger adversary observes a sequence of releases for a user and decodes the most likely hidden node sequence by Viterbi, combining the per-report emission model with the Markov mobility prior. Privacy is the mean tracking error AE_{traj} (metres) and the per-step re-identification rate. This is the threat temporal mechanisms target, and the one that independent per-report noise fails to resist because temporal correlation lets the adversary average noise out [12].

4.3 Why a Single Scale Is Inadequate

The two adversaries induce different rankings (Section 8, Fig. 2): a mechanism strong against one can be weak against the other, so any single-scale privacy comparison mis-ranks mechanisms defending different threats. This is the central methodological point of the paper.

1. <https://github.com/Pushkal-Gupta/Graph-based-Location-Privacy-IoT>

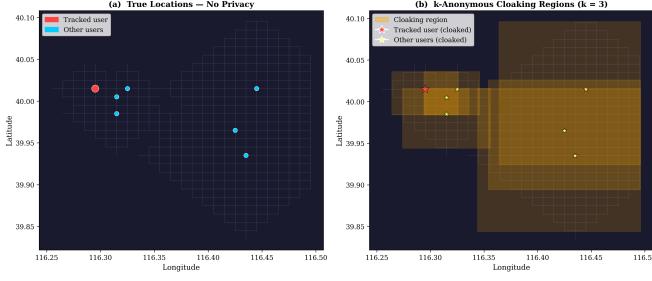


Fig. 1. Spatial cloaking on the Beijing graph: true locations (left) are replaced by the medoid of a connected region containing at least k users (right).

Algorithm 1 DA-Hybrid per-report anonymisation

Require: snapshot S , target k , base budget ϵ , size cap R

- 1: for each user u at node v do
- 2: $\rho \leftarrow$ local density of v in S
- 3: grow region $\mathcal{R}$ by BFS until $|\text{users}(\mathcal{R})| \geq k$ or $|\mathcal{R}| \geq R(\rho)$
- 4: if $|\text{users}(\mathcal{R})| \geq k$ then
- 5: report $\text{medoid}(\mathcal{R})$ $\{k\text{-anonymity branch}\}$
- 6: else
- 7: $\epsilon' \leftarrow$ density-adaptive budget ($\downarrow$ in sparse regions)
- 8: add $\text{Lap}(\Delta/\epsilon')$ noise; project to nearest node $\{\text{DP fallback}\}$
- 9: end if
- 10: end for

5 Privacy Mechanisms

We evaluate five representative classical mechanisms spanning three families, plus our DA-Hybrid (Section 6).

Graph k -anonymity grows a connected region by breadth-first search until it covers $\geq k$ users, and reports the region medoid [6], [7] (Fig. 1). Coordinate differential privacy adds planar Laplace noise $\text{Lap}(\Delta/\epsilon)$ to each coordinate [8]; the output may fall off the road network. Graph-constrained DP projects the noisy point to the nearest graph node [10], restoring on-graph validity. Density-aware adaptive k -anonymity selects k per node from local density percentiles [11]. Temporal cloaking delays reports until k users share a temporal window and releases the group medoid [2].

6 DA-Hybrid: Density-Adaptive Hybrid Privacy

The comparison reveals a gap: k -anonymity attains low error where users are dense but loses availability in sparse areas, while DP guarantees availability at an error premium. DA-Hybrid closes this gap with a per-report, density-driven controller (Algorithm 1).

For a user at node v , let $\rho(v)$ be the local density (users at v plus its one-hop neighbours). The mechanism (i) attempts bounded graph k -anonymity, growing a region until it covers $\geq k$ users or reaches a density-dependent size cap; if k is met it reports the medoid (low error, on-graph, k -satisfied). Otherwise (ii) it falls back to graph-constrained DP with a density-adaptive budget— ϵ is reduced in sparse regions, precisely where anonymity is hardest and most needed—and projects to the nearest node.

Because the fallback never fails, availability is 100 % by construction; because dense reports are served by cloaking,

error stays low where density permits; and every output is a valid road-network node carrying a per-report guarantee of one of two well-understood types. The cost matches the underlying primitives (bounded BFS, or $O(1)$ noise plus nearest-node projection).

7 Experimental Setup

All mechanisms run on the same graph and data with a shared distance layer. We sweep $k \in \{2, \dots, 6\}$, $\epsilon \in \{0.1, 0.5, 1, 2, 5\}$, and time windows $\{60, 300, 600, 1200\}$ s; unless noted, results use the representative 10-minute window. Privacy is measured by the two adversaries of Section 4; availability is the fraction of reports served with their guarantee; utility is the spatial (graph) location error. The adversary prior and Markov mobility model are estimated once per dataset and shared. On the real graph a single dense distance matrix is shared by every mechanism and the adversary, keeping the exact Bayes estimator tractable.

Energy model. Per-report energy is $E = E_{\text{radio}} + E_{\text{comp}} + E_{\text{retrans}}$, with $E_{\text{radio}} = P_{\text{tx}} t_{\text{tx}}$. Rather than fix a single radio, we sweep BLE, Zigbee, LoRa, NB-IoT, and LTE-M (Section 8.4) so that radio-dependent conclusions are identified as such.

8 Results

8.1 Adversary-Grounded Comparison

Table 1 reports all mechanisms under both adversaries, availability, spatial error, on-graph validity, and per-report energy under two radios (GeoLife, grid, 10 min). Three findings stand out. (1) Under a real Bayesian adversary, k -anonymity is far more private than its $[0, 1]$ score implied—snapshot error 5193 m at $k=3$ versus 1651 m for DP at $\epsilon=1$ —whereas the old score rated them comparable. (2) The trajectory adversary reorders mechanisms relative to the snapshot adversary (Fig. 2): density-aware adaptation and temporal cloaking swap rank, and temporal cloaking—most private under the snapshot attack—is markedly less so under tracking. (3) DA-Hybrid attains the lowest error among the 100 %-availability mechanisms (1117 m vs. 1569 m for DP and 1890 m for graph-constrained DP) while remaining on-graph (Fig. 3).

8.2 Availability

Fig. 4 shows availability varies dramatically on GeoLife: both DP variants and DA-Hybrid serve all users, k -anonymity 74.5 %, density-aware 78 %, and temporal cloaking only 18.5 %. As we show next, however, these figures are dataset-specific.

8.3 Real Road Network and Second Dataset

We repeat the evaluation on the real OSM graph with GeoLife map-matched to it, and again on the denser T-Drive taxis (Table 2, Fig. 5). Two robust conclusions emerge. Rankings hold across topology: under the adversary-privacy metric the ordering on the real graph is nearly identical to the grid—temporal cloaking most private, DA-Hybrid most utility-favouring—so relative rankings are shown, not merely asserted, to survive the grid abstraction; only the DP/graph-constrained-DP pair exchanges places. Availability is dataset-density-driven, not mechanism-intrinsic: on sparse GeoLife,

TABLE 1

Unified comparison under adversary-grounded privacy (expected Bayesian-adversary error in metres; higher is more private) for the snapshot and trajectory threat models. GeoLife, 30×30 grid, window = 10 min.

Mechanism	Snap. AE (m)	Traj. AE (m)	Avail.	Loc. Err (m)	On-graph	E_{LoRa} (mJ)	E_{BLE} (mJ)
k -Anonymity	5193	5141	74.5%	1189	yes	5.57	0.62
Differential Privacy	1651	1893	100%	1569	no	5.05	0.10
Graph-Constr. DP	1829	1915	100%	1890	yes	5.16	0.21
Density-Aware	4256	5563	78.2%	2427	yes	5.78	0.83
Temporal Cloaking	6376	5037	18.5%	5758	yes	5.05	0.10
DA-Hybrid (ours)	1441	1707	100%	1117	yes	5.10	0.15

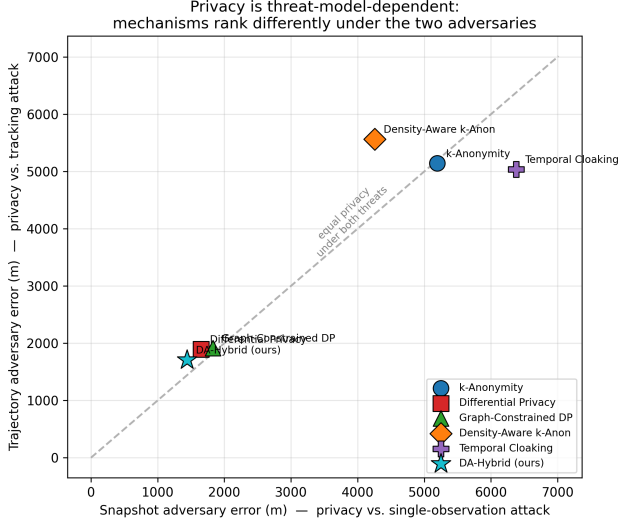


Fig. 2. Privacy is threat-model dependent. Each mechanism’s expected adversary error under the snapshot attack (x) and the trajectory attack (y). Points off the $y=x$ diagonal are protected asymmetrically; density-aware and temporal cloaking swap rank between the two threats.

TABLE 2

Location error (m) / availability across the grid abstraction, the real OSM graph under GeoLife, and the real graph under T-Drive (10 min).

Mechanism	GeoLife/grid	GeoLife/real	T-Drive/real
k -Anonymity	1189 / 74%	2517 / 77%	107 / 100%
Differential Privacy	1569 / 100%	414 / 100%	454 / 100%
Graph-Constr. DP	1890 / 100%	933 / 100%	809 / 100%
Density-Aware	2427 / 78%	2040 / 72%	128 / 100%
Temporal Cloaking	5758 / 19%	4475 / 90%	6460 / 100%
DA-Hybrid	1117 / 100%	664 / 100%	62 / 100%

k -anonymity (74.5%) and temporal cloaking (18.5%) suffer denial of service, but on dense T-Drive every mechanism reaches 100% availability. Absolute spatial error scales with graph granularity—the Laplace scale is calibrated to the median edge length, which is shorter on the real graph—so error is compared within, not across, topologies.

8.4 Energy-Model Sensitivity

The claim that radio transmission dominates per-report energy is not universal: it depends on the radio (Fig. 6). Sweeping BLE (0.05 mJ) through NB-IoT/LTE-M (60–75 mJ), the

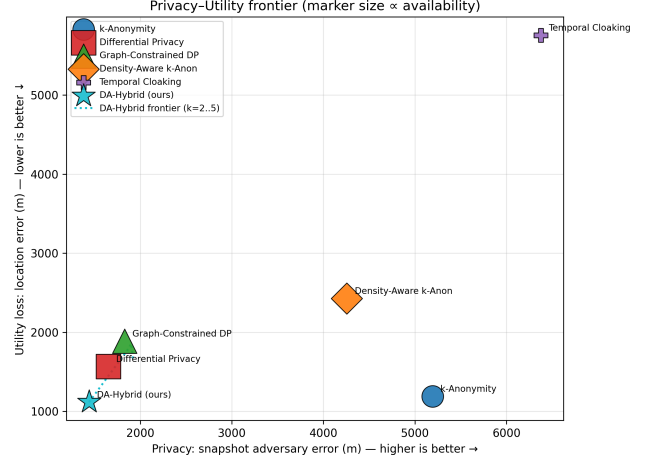


Fig. 3. Privacy (snapshot adversary error, higher is better) versus utility loss (location error, lower is better); marker size $\propto$ availability. No mechanism dominates: k -anonymity reaches high privacy but only 74.5% availability, while DA-Hybrid gives the lowest error among the 100%-availability mechanisms.

Service Availability Analysis

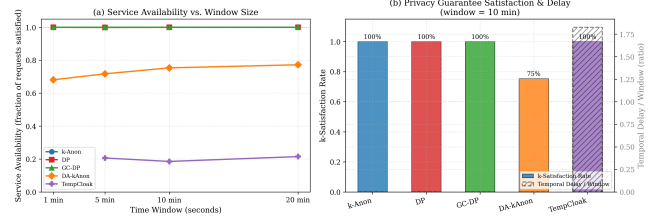


Fig. 4. Service availability on GeoLife (grid). DP variants and DA-Hybrid reach 100%; temporal cloaking is critically low at 18.5%.

radio share ranges from 6–50% under BLE—where the BFS-based mechanisms’ computation dominates—to 99% under cellular IoT. The “radio dominates” conclusion is therefore scoped to LPWAN/cellular deployments; for short-range radios, algorithm computation is a first-order energy factor.

9 Discussion

Deployment guidance. The right mechanism depends on the binding constraint. Where denial of service is unacceptable, use DA-Hybrid or a DP variant (100% availability); DA-Hybrid additionally keeps outputs on-graph and gives the lowest error among these. Where the strongest privacy against a single-observation attacker is paramount and some service

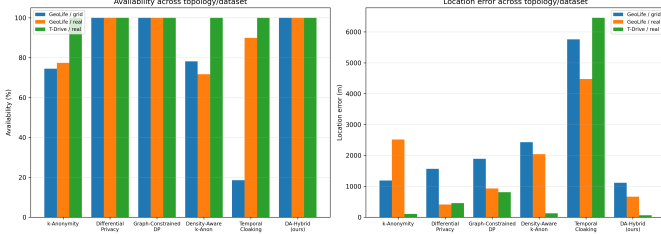


Fig. 5. Availability (left) and location error (right) across the grid, the real OSM graph (GeoLife), and the real graph under T-Drive. Availability rises to 100% for all mechanisms on the dense dataset; DA-Hybrid holds the lowest error throughout.

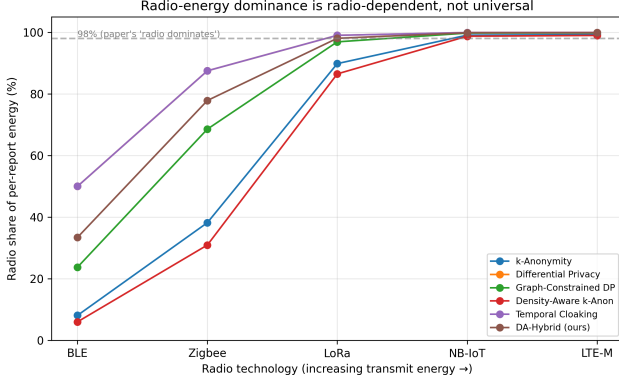


Fig. 6. Radio share of per-report energy across radio technologies. Radio dominance holds for LoRa/NB-IoT/LTE-M but not for BLE, where computation is comparable or larger for BFS-based mechanisms.

loss is tolerable, k -anonymity or temporal cloaking rank highest—but only against the snapshot adversary; under tracking their advantage shrinks, so the choice must be made against the relevant threat. Availability requirements should be read against deployment density: on dense fleets (T-Drive-like) all mechanisms serve every user, so availability ceases to discriminate.

Limitations. Absolute error is not comparable across topologies because the Laplace scale is graph-granularity dependent; our cross-topology conclusions therefore rest on adversary-privacy and availability. The energy model is analytical, suited to ranking rather than absolute prediction, and we report its radio sensitivity explicitly. The mobility prior is first-order Markov; richer priors would strengthen the trajectory adversary and can only lower the reported privacy.

10 Conclusion

We replaced ad-hoc privacy scores with an adversary-grounded metric evaluated under two threat models, showing that a single privacy scale mis-ranks mechanisms that defend different threats. Using this metric we introduced DA-Hybrid, a density-adaptive mechanism with 100% availability, on-graph outputs, and the lowest error among full-availability mechanisms. Across a real Beijing road network and two datasets, adversary-privacy rankings are stable while service availability is shown to be dataset-density-driven rather than mechanism-intrinsic. The framework is released for reproducible extension.

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